

Mining Social-Network Graphs

By

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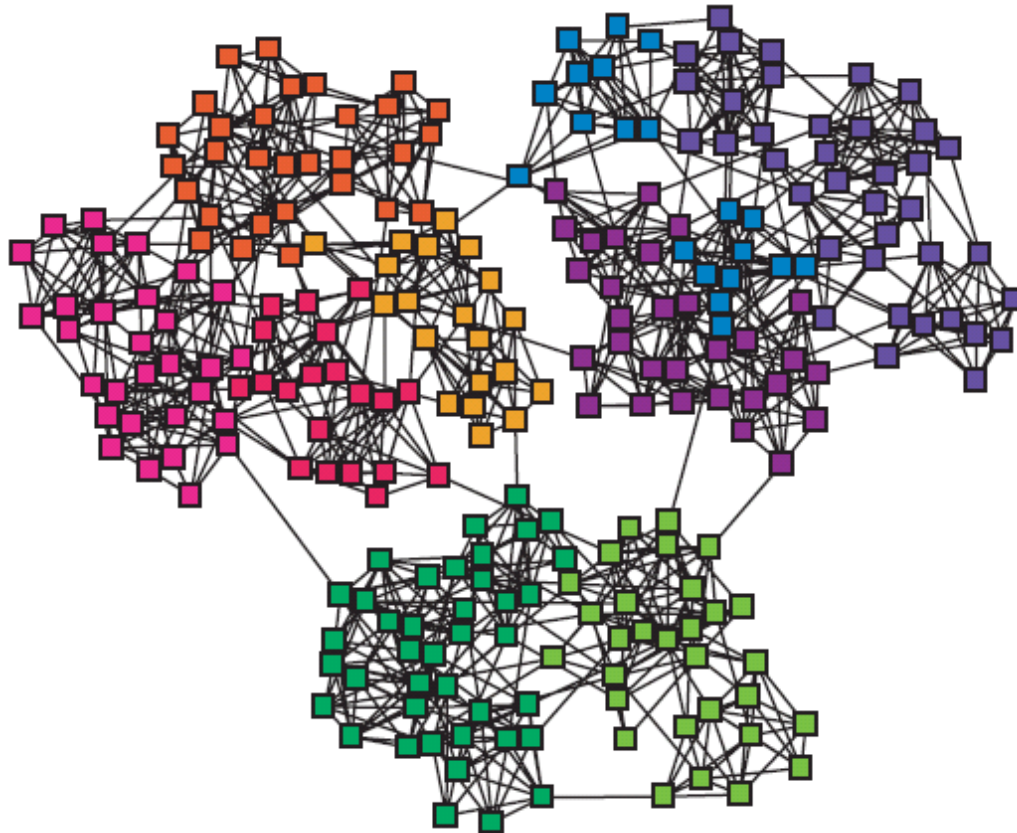
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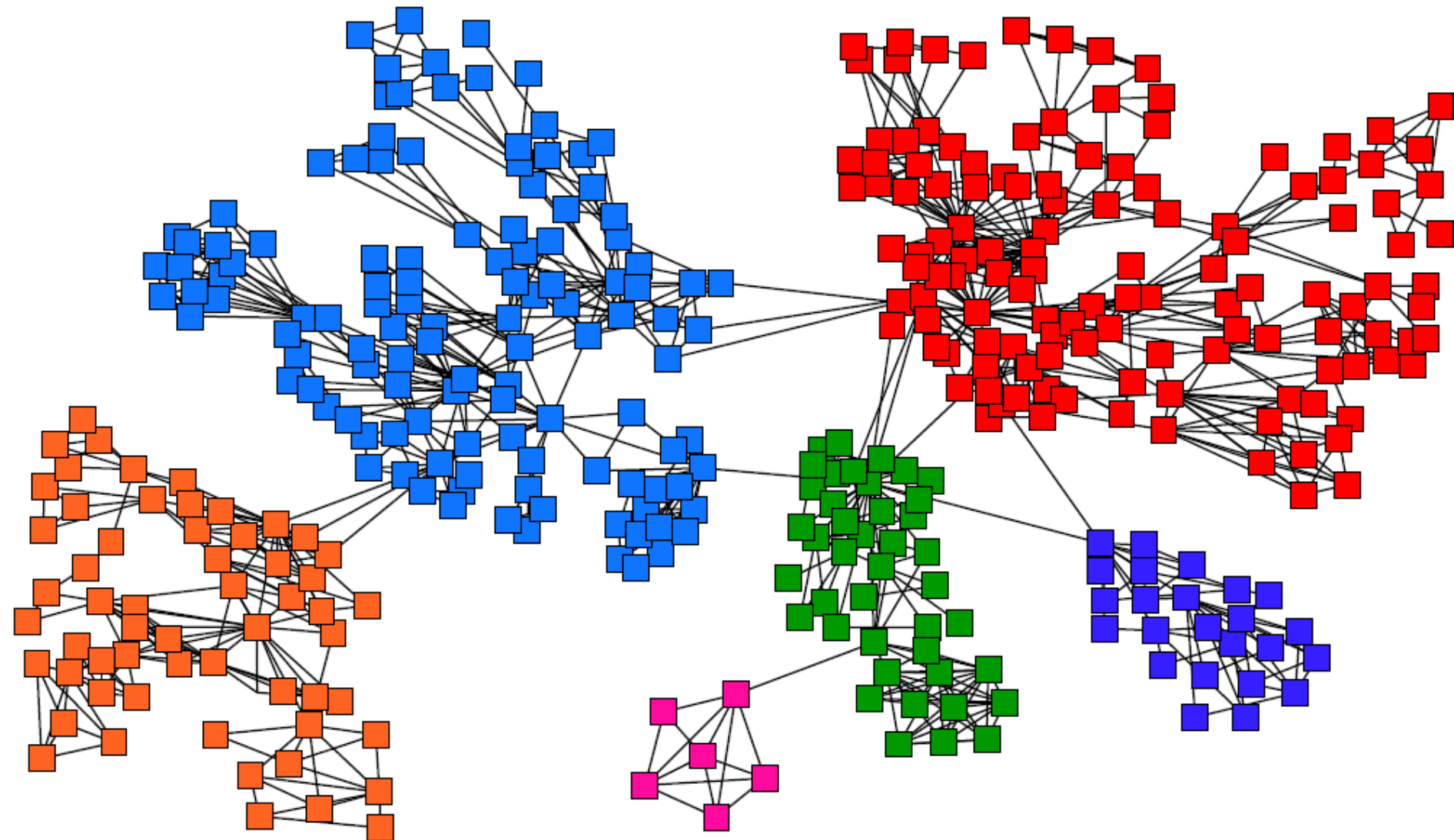
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Networks & Communities

- We often think of networks being organized into **modules, cluster, communities:**

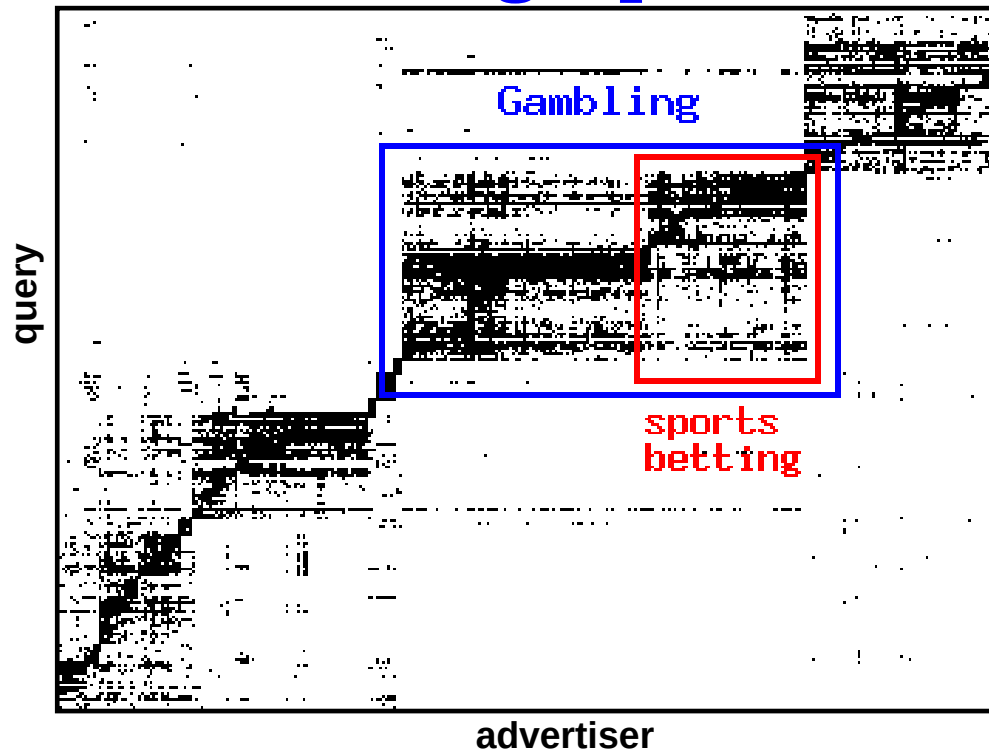


Goal: Find Densely Linked Clusters



Micro-Markets in Sponsored Search

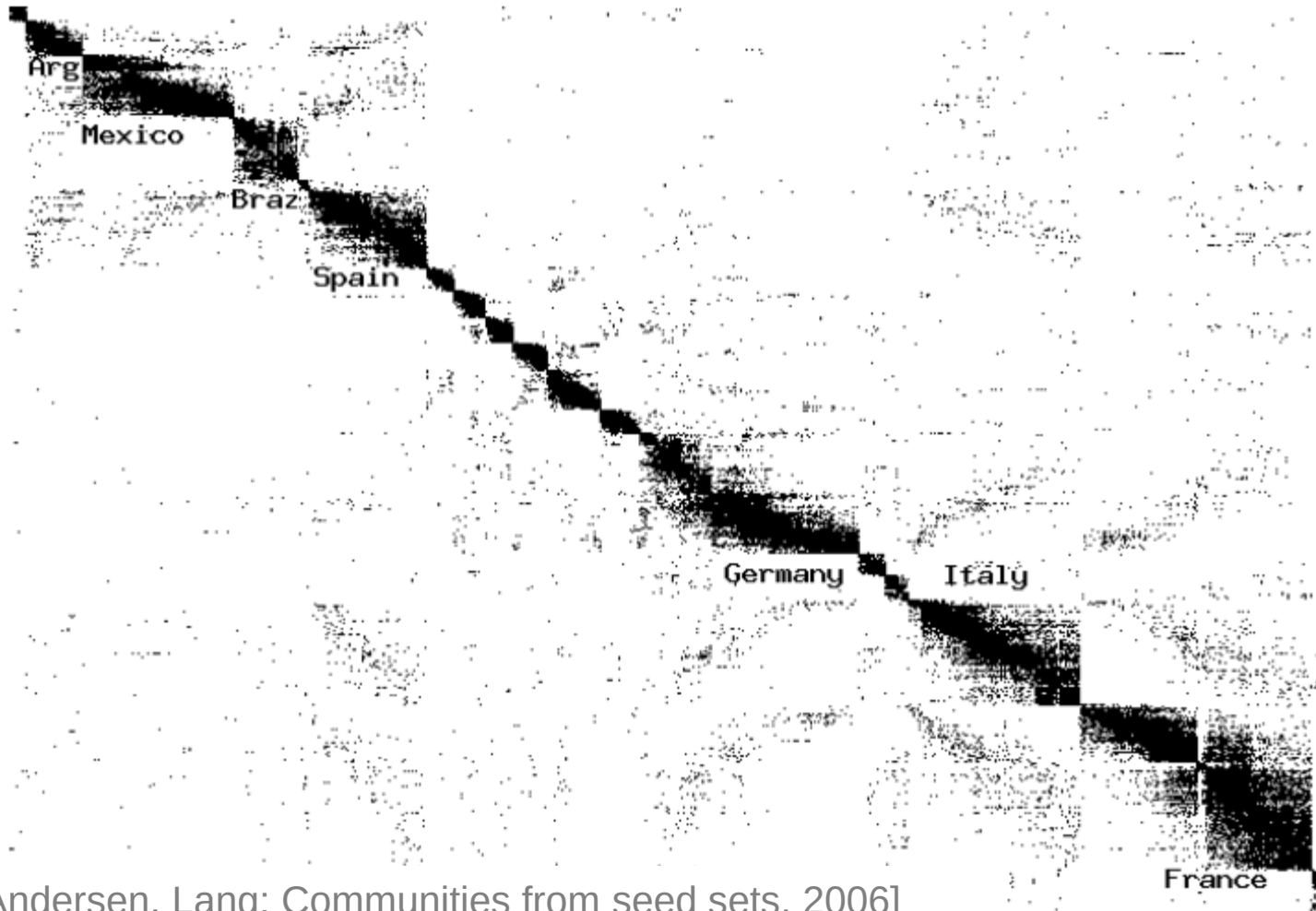
- Find micro-markets by partitioning the query-to-advertiser graph:



[Andersen, Lang: Communities from seed sets, 2006]

Movies and Actors

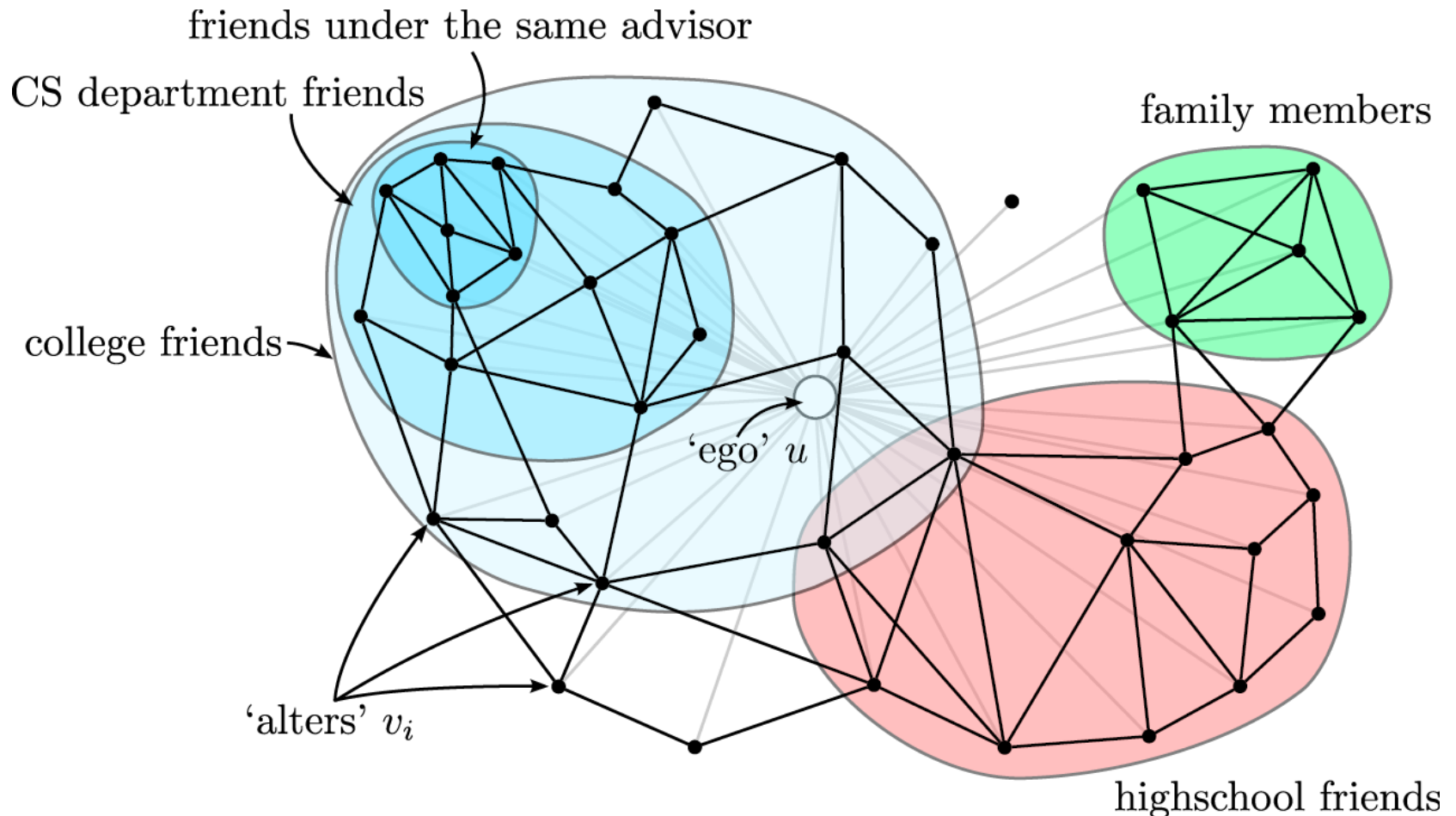
- **Clusters in Movies-to-Actors graph:**

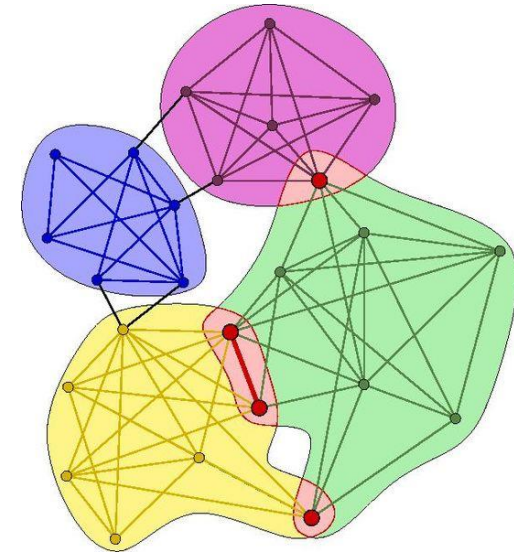
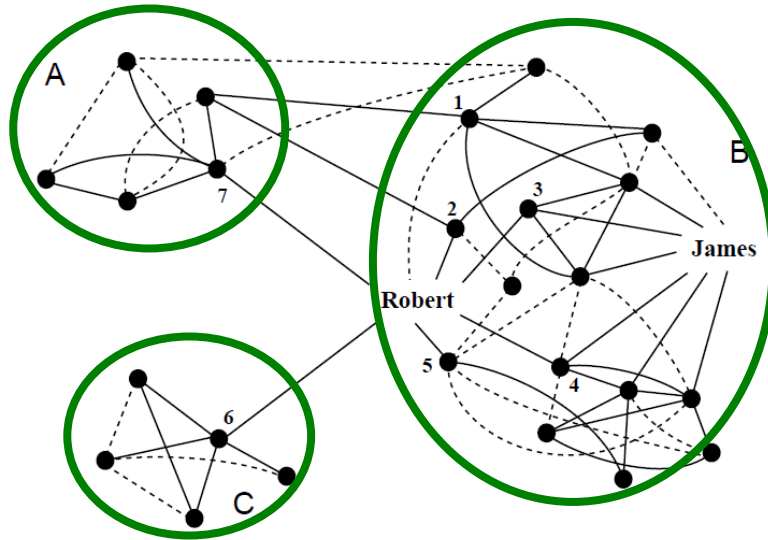


[Andersen, Lang: Communities from seed sets, 2006]

Twitter & Facebook

- **Discovering social circles, circles of trust:**





How to find communities?

COMMUNITY DETECTION

We will work with **undirected** (unweighted) networks

Types of social networks

- **Collaboration Graph**
 - co-authorships among scientists and co-appearance in movies
- **Who-Talks-to-Whom Graphs**
- **Information Linkage Graphs**
- **Heterogeneous Social Network**
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Thank You